

Samiksha Chauhan

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PROFESSIONAL SUMMARY

Results-driven Data Analyst and ML Engineer with hands-on experience building end-to-end machine learning pipelines, RAG-based AI systems, and interactive BI dashboards. Proficient in Python, SQL, Power BI, and modern MLOps practices. Demonstrated ability to engineer production-grade solutions—from predictive maintenance models achieving 99% accuracy to containerized LLM applications—translating complex data into actionable business insights.

EDUCATION

GLA University, Mathura — B.Tech, Computer Science (AI/ML)

2022 – 2026

CGPA: 7.04

Milton Public School, Agra

Intermediate: 84.8% • High School: 87%

TRAINING

Advanced Data Science – Value Added Course (Remote Intern)

Jun – Jul 2023

- Performed structured data cleaning, preprocessing, and EDA workflows using Python (Pandas, NumPy).
- Built analytical workflows and statistical visualizations using Scikit-learn, Matplotlib, and Seaborn.

PROJECTS

End-to-End Predictive Maintenance AI System

Python, XGBoost, Scikit-learn, Streamlit

- Engineered a full MLOps pipeline on the UCI AI4I 2020 dataset to predict industrial machine failures from telemetry signals including rotational speed, torque, and operating temperature.
- Applied Label Encoding and stratified train-test splits; enforced strict DataFrame segregation to eliminate target data leakage from five concurrent failure-mode columns (TWF, HDF, PWF, OSF, RNF).
- Trained an XGBoost classifier achieving ~99% global accuracy and a 0.96 ROC-AUC score, with robust resistance to feature scaling.
- Deployed the model via a Streamlit web interface with an abstracted UI-to-inference mapping layer to resolve feature name mismatches between form labels and trained column names.
- Resolved Windows MAX_PATH build failures by replacing JupyterLab with a lightweight nbformat-based pure-Python notebook abstraction, unblocking the CI pipeline.

Enterprise RAG Document Assistant

Python, FastAPI, LangChain, ChromaDB, Docker, LLaMA-3

- Built a full-stack Retrieval-Augmented Generation (RAG) application enabling private, hallucination-free Q&A over enterprise PDF documents using a microservices architecture.
- Designed a 6-stage RAG pipeline: PDF ingestion via PyMuPDF → LangChain recursive text chunking (1000 tokens, 200 overlap) → SentenceTransformer embeddings → ChromaDB vector storage → cosine similarity retrieval → LLaMA-3.3-70B response synthesis via Groq Cloud API.
- Resolved context window fragmentation across multi-document queries by scaling retrieval scope from k=3 to k=5 and upgrading the inference model from LLaMA-3.1-8B to LLaMA-3.3-70B-Versatile, eliminating false 'cannot answer' responses.
- Containerized the entire platform using Docker & Docker Compose, decoupling the FastAPI backend from the Streamlit frontend over an isolated virtual network for reproducible, portable deployments.
- Debugged and resolved a PyTorch version incompatibility (2.0.1 vs. required 2.4+) that crashed the SentenceTransformers embedding phase by upgrading core ML dependencies.

TECHNICAL SKILLS

Analytics & BI: Power BI, Excel (Pivot Tables, XLOOKUP, Conditional Formatting)

Programming & Querying: Python, SQL

ML & AI Frameworks: Scikit-learn, XGBoost, LangChain, SentenceTransformers, Streamlit, FastAPI

Libraries: Pandas, NumPy, Matplotlib, Seaborn, PyMuPDF

Databases & DevOps: ChromaDB, SQLite, MySQL, Docker, Docker Compose, Git, Jupyter Notebook

Core Concepts: Data Cleaning, EDA, Feature Engineering, MLOps, RAG Architecture, Vector Embeddings, Dashboard Development, KPI Analysis, Predictive Modeling

CERTIFICATIONS

- SQL (Basic to Advanced) – HackerRank
- Data Science Simulation – British Airways (Forage)
- Azure AI NLP – Microsoft
- Azure Document AI – Microsoft